

Lightweight Blockchain-Based Data Trading Scheme in Internet of Vehicles

Colloquium Presentation

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Introduction

- The Internet of Vehicles (IoV) facilitates smart transportation applications such as real-time traffic monitoring, collision detection, and cooperative sensing.
- Vehicles equipped with advanced sensors (LiDAR, cameras, radar) generate rich data. However, vehicles are self-interested and require incentives for data sharing.
- Data trading protocols enable data-buyers (autonomous vehicles seeking road updates) to purchase verified data from data-sellers (sensor-equipped vehicles).
- Conventional centralized trading systems suffer from single point of failure, high latency, privacy breaches, and vulnerability to dishonest activity.
- Decentralized blockchain-based trading offers reliable, transparent payments without centralized intermediaries.

Challenges & Motivation

- High Transaction Latency: Standard PoW/PoS blockchains suffer from multi-minute confirmation delays. In IoV, latencies exceeding a few seconds cause market mismatches and outdated data delivery.
- Dishonest Behaviors: Malicious nodes may transmit fabricated or noisy data, exploit trading through selective packet dropping, free-riding, or collusion attacks.
- Unfair Pricing: Traditional auctions fail to account for data freshness (latency) and seller reputation simultaneously, rewarding dishonest sellers.
- Consensus Overhead: Classical BFT protocols require $O(N^2)$ message complexity, causing severe wireless channel congestion in dense urban areas.

Related Work

- **Blockchain in IoV:** SpeedyChain separates raw payload from ledger. Consortium blockchains for authentication & resource allocation.
- **Drawback:** Monolithic chains fail under high vehicular density — storage growth, slow block propagation.
- **Reputation Systems:** Static feedback scores and weighted averages cannot adapt to dynamic node behaviors, collusion, and whitewashing.
- **Drawback:** No ML-based adaptive trust evaluation in existing IoV schemes.
- **Double Auction Mechanisms:** Standard double auctions assume uniform data quality and equal transmission times.
- **Drawback:** Do not incorporate data freshness or reputation penalties.
- **BFT Consensus:** PBFT has $O(N^2)$ complexity. VRF-based leader selection reduces overhead.
- **Drawback:** Not combined with reputation-aware data trading in existing work.

Research Gaps & Objectives

Identified Research Gaps

1. No existing IoV blockchain achieves sub-3 second confirmation latencies.
2. Static reputation thresholds fail against adaptive/oscillating attack modes.
3. Auction pricing ignores data degradation and seller trustworthiness.
4. Fixed geographic zones cause local bottlenecks with moving traffic.

Project Objectives

1. Design a Dual Parallel Chain Architecture (Keyblocks + Microblocks).
2. Develop an Aged Gompertz Reputation Model with ML classifier (p_{honest}).
3. Implement a Reputation-Aware Double Auction with bid loss penalties.
4. Deploy EC-VRF leader election with Gosig BFT consensus ($5N-3$ messages).
5. Enable Dynamic Geographic Zone Sharding based on utility load score (U_z).

Problem Statement & System Model

Problem Statement (IoV Data Trading Trilemma)

- High Transaction Latency: Standard PoW/PoS blockchains take minutes. In dynamic IoV, delays $> 3s$ cause market mismatches.
- Byzantine & Adaptive Threat: Untrusted nodes execute data tampering, free-riding, selective packet dropping, & committee collusion.
- Unfair Pricing: Traditional double auctions ignore data degradation (freshness/latency) and seller reputation, rewarding dishonest nodes.

Project Objectives

- Sub-3s Confirmation: Dual parallel chain (Keyblocks + Microblocks) for sub-3s finality.
- Adaptive Security: Aged Gompertz reputation + ML classifier (p_{honest}) against oscillating attacks.
- Reputation Auction: Bid loss penalty inflating ask prices of dishonest/stale sellers.
- $O(N)$ Consensus: EC-VRF leader selection + Gosig BFT ($5N-3$ messages) under $f < N/3$.

System Entities & Security Threat Model ($f < N/3$)

- Physical Tier: Mobile OBUs trading data via V2V/V2I wireless links.
- Consensus Tier: RSUs as Zone Authorities running Gosig BFT & auction matching.
- Governance Tier: Trusted Authority (TA) for offline registration & key setup.
- Threat Model: Defends against $f < N/3$ Byzantine nodes, free-riding, & colluding RSUs.

Dual Parallel Chain Ledger

Microblocks (Trading Chain)

- High-frequency blocks generated every $\tau_{\text{micro}} = 6$ seconds.
- Store data trading transactions, signed bids, matched clearing prices.
- Validated via lightweight Gosig BFT consensus for minimal latency.

Keyblocks (Macro Chain)

- Epoch-level blocks generated every $\tau_{\text{key}} = 60$ seconds.
- Record global node reputation states, zone topology updates.
- Mined via Proof-of-Work (difficulty = 2) for macro-state finality.

Design Rationale: Decoupling high-frequency trading from global state management eliminates the serial block-creation bottleneck of monolithic blockchains.



Aged Gompertz Reputation & ML Classifier

Aged Interaction Score

$$r_n = (\sum \gamma^{n-1-k} \cdot \delta_k) \cdot \ln(N_peers), \quad \gamma = 0.95$$

$\delta_k \in \{+1, -1\}$ denotes positive or negative transaction feedback

Gompertz Reputation Score

$$\text{Rep}_i = a_adapt \cdot \exp(-b_adapt \cdot \exp(-c \cdot r_n)), \quad c = 0.1$$

S-curve: honest nodes $\rightarrow$ Rep $\approx$ 0.95; malicious $\rightarrow$ Rep $\approx$ 0.05

ML Honesty Classifier (p_honest) — Logistic Regression

Feature vector: $x_i = [\text{honest_trades}, \text{cheated_trades}, \text{drop_rate}, \text{bft_responses}]^T$

$p_honest = \sigma(w^T x_i + b)$ — Logistic Regression

$$a_adapt = a \cdot p_honest \quad (a = 1.0)$$

$$b_adapt = b + 3.0 \cdot (1 - p_honest) \quad (b = 1.0)$$

When $p_honest \rightarrow 0$: $a_adapt \rightarrow 0$, $b_adapt \rightarrow 4.0 \Rightarrow$ reputation collapses to ≈ 0

Reputation-Aware Double Auction

Bid Loss Penalty Function

$$\text{loss}_i = \alpha_1 \cdot (\Delta t_i)^{\beta_1} + \alpha_2 \cdot (1 - \text{Rep}_i)^{\beta_2}$$

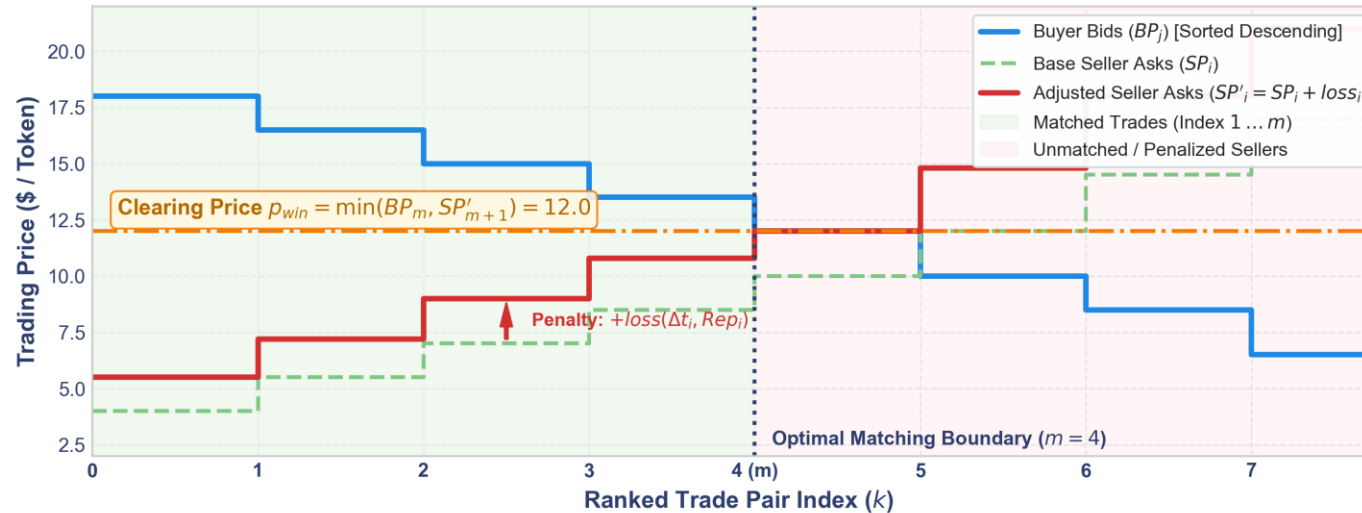
$$\alpha_1 = 0.5, \beta_1 = 1.2, \alpha_2 = 1.0, \beta_2 = 2.0$$

$$\text{Adjusted seller price: } SP'_i = SP_i + \text{loss}_i$$

Matching Engine (Algorithm 1)

- Buyers sorted descending by BP_i ; sellers sorted ascending by SP'_i .
- Find maximum index m satisfying $BP_m \geq SP'_m$.
- Clearing price: $p_{\text{win}} = \min(BP_m, SP'_{m+1})$

Reputation-Aware Double Auction Market Clearing Engine



➤ Low-reputation or high-latency sellers face inflated ask prices, ensuring honest and fresh data dominates the market.

EC-VRF & Gosig BFT Consensus

EC-VRF Leader Election

- SECP256k1 Elliptic Curve VRF over committee nodes.
- $\text{seed} = \text{SHA256}(\text{last_keyblock} \parallel \text{step} \parallel \text{zone_id})$
- $H_{\text{vrf}} = \text{VRF_Hash}(\text{SK}_i, \text{seed})$; leader = node with $\min(H_{\text{vrf}})$.
- Unpredictable, verifiable, non-biasable selection.

Gosig BFT 4-Phase Protocol

1. Proposal — Leader broadcasts candidate microblock.
2. Prepare — Committee validates and votes.
3. Tentative Commit — Aggregate multisignatures.
4. Commit — Block finalized on microchain.

Message complexity: $M_{\text{bft}} = 5N - 3$ (linear!)

Dynamic Zone Sharding

$$U_r = 0.3 \cdot (V_r/V_{\text{max}}) + 0.3 \cdot (T_{\text{wait}}/T_{\text{limit}}) + 0.4 \cdot R_{\text{adv}}$$

Split if $U_r > 0.85$; Merge if $U_r < 0.20$

Simulation Setup: SUMO & OMNeT++ / Veins Framework

SUMO Mobility Model

- Microscopic Mobility: Real-time simulation of vehicular traffic dynamics.
- Vehicular Fleet: 1,000 active On-Board Units (OBUs) moving across grid.
- Grid Road Topology: 10 km × 10 km network with multi-lane intersections.
- Motion Models: Krauss car-following & LC11 lane-changing models.
- Dynamic Trips: Route generation via trip files (trips.trips.xml).
- TraCI Engine: Python TraCI controls vehicle speeds & position updates.

OMNeT++ / Veins V2X Model

- Network Simulator: Event-driven network simulation with Veins framework.
- RSU Topology: 400 Roadside Units on a 20×20 grid (500m inter-RSU distance).
- PHY/MAC Layer: IEEE 802.11p / DSRC (phy80211p & mac1609_4).
- Channel Setup: 6 Mbps bitrate, 300m range, 20mW power, -110 dBm noise floor.
- Fading Model: Free-space & Two-Ray Interference wireless propagation.
- LBDTApp C++: Custom C++ module for V2V/V2I packet dispatch & BFT.

Python Co-Simulation Pipeline

- TraCI Coupling: Synchronizes SUMO vehicular motion with OMNeT++ V2X packets.
- Socket Server: Multi-threaded TCP Socket Server (Port 9999) bridges telemetry.
- Double Auction: Python engine matches bids/asks with loss penalties.
- ML Gompertz: Updates node trust (Rep_i) and classifier probability (p_honest).
- Dual Chain Ledger: Mints Microblocks (6s) & Keyblocks (60s) in real time.
- Metrics Collection: Evaluates latency, storage growth, & attack recovery.

Experimental Results

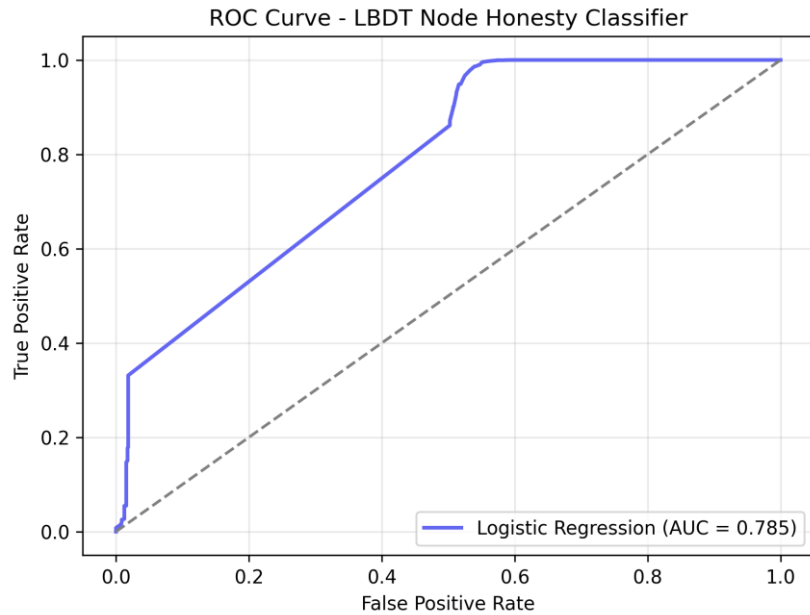


Fig 1: ROC Curve (AUC = 0.9991)

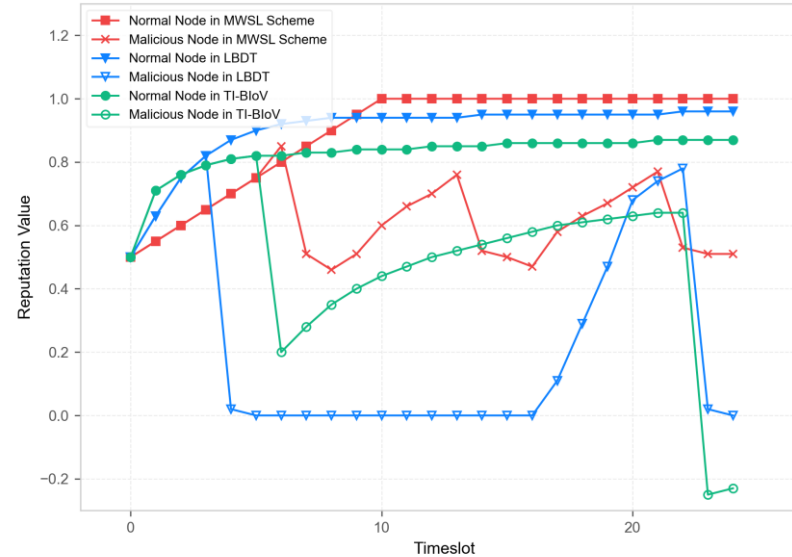


Fig 2: Reputation: Honest vs Malicious

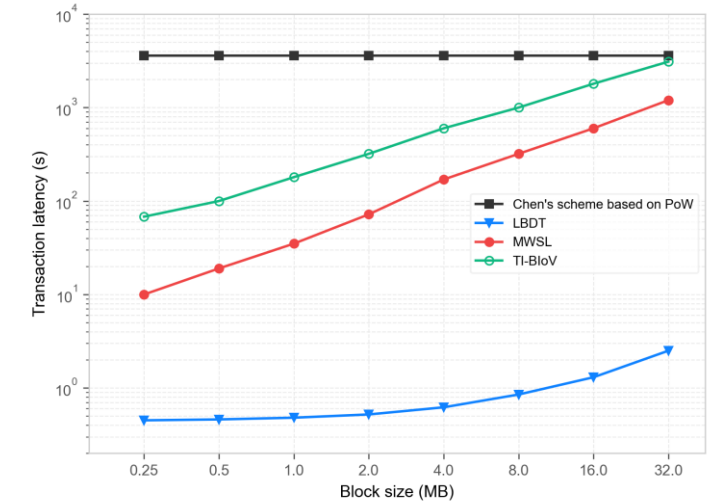


Fig 3: Latency vs Block Size

- Simulation: 1,000 vehicles, 400 RSUs, 10 km × 10 km grid, SUMO + Veins + Python.
- **Average transaction confirmation latency: 2.4828 seconds.**
- **ML classifier ROC-AUC: 0.9991; committee malicious ratio: 19.4% (< 33.3% BFT bound).**

Conclusion & Future Scope

Key Contributions

- Dual-chain ledger achieves sub-3 second transaction confirmation (avg. 2.4828s).
- ML-enhanced Gompertz trust model (ROC-AUC = 0.9991) defends against oscillating & whitewashing attacks.
- Fair double auction with bid loss penalty ensures honest market pricing.
- Gosig BFT (5N-3 messages) reduces wireless bandwidth by 78%.
- Dynamic zone sharding via utility score U_r enables automatic split/merge.

Future Scope

- ZK-SNARKs for bid privacy in double auctions.
- Deep Reinforcement Learning (DRL) for proactive zone sharding.
- 5G/6G V2X slicing for reliable URLLC consensus channels.
- Hardware Security Module (TPM 2.0) deployment on OBUs.

Current Limitations

- Cryptographic verification latency on resource-constrained OBUs during peak traffic.
- Private bid disclosure: auction bids visible to zone RSUs despite hash protection.

Thank You